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ITA0303-MOBILE COMPUTING

4. Mobile IP Application Design for Smart Healthcare (BL4 – Analyze)

Design a Smart Healthcare Mobile IP system using Figma that enables doctors, ambulances, hospitals,

and patients to remain connected while moving across different wireless networks.

Analyze:

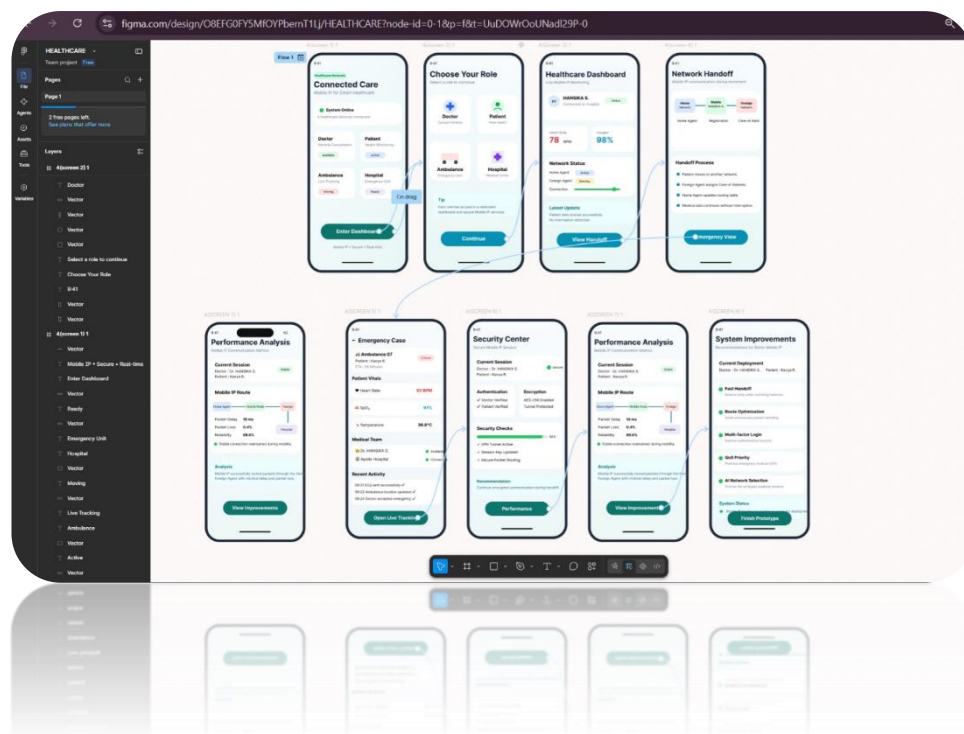
- ❑ How Mobile IP maintains uninterrupted communication.
- ❑ Challenges during handoff between networks.
- ❑ Security vulnerabilities during mobility.
- ❑ Possible design improvements to increase reliability.

Provide navigation between screens demonstrating different mobility scenarios.

Deliverables:

- ❑ Interactive prototype.
- ❑ GitHub repository containing design files, exported prototype, and documentation.

PROTOTYPE:



FIGMA LINK:

<https://www.figma.com/design/O8EFG0FY5MfOYPbernT1Lj/HEALTHCARE?node-id=6-157&t=ltObzaYgjZgXbQqd-0>

5. Wireless Network Decision Dashboard Using GitHub Project (BL4 – Analyze)

Create a web-based dashboard (using HTML/CSS/JavaScript, React, or another programming framework) or an interactive Figma dashboard that helps network engineers determine whether a given

scenario should use Mobile IP, Cellular Networks, or Ad Hoc Networks.

The dashboard should analyze multiple parameters such as:

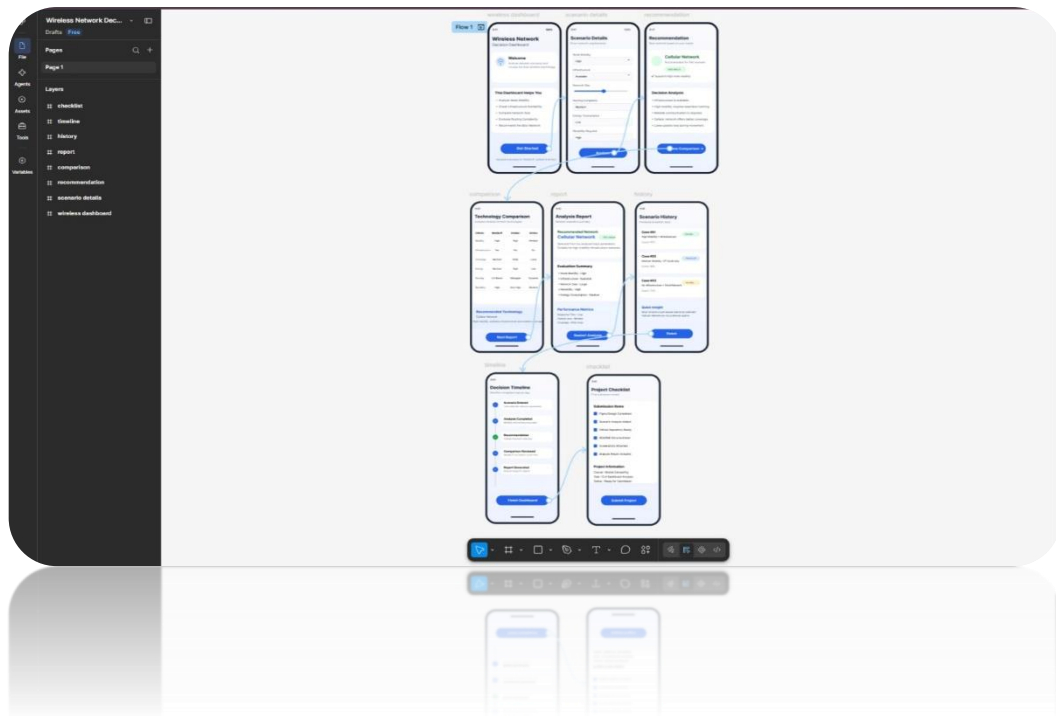
- ☐ **Node mobility.**
- ☐ **Infrastructure availability.**
- ☐ **Network size.**
- ☐ **Routing complexity.**
- ☐ **Energy consumption.**
- ☐ **Reliability requirements.**

The system should justify its recommendation based on the analyzed parameters.

Deliverables:

- ☐ **Source code or Figma project.**
- ☐ **GitHub repository with complete code/design, README, screenshots, and analysis report.**

PROTOTYPE:



FIGMA LINK:

<https://www.figma.com/design/z9dU4gHRBgFm5FFQFu6W5J/Wireless-Network-Decision-Dashboard?node-id=0-1&p=f&t=4bzW6XEC0NA2au7N-0>

3. Routing Protocol Visualization and Analysis (BL4 – Analyze)

Develop an interactive visualization using Figma, Mermaid.js, D3.js, or any visualization framework

illustrating the operation of one Proactive Routing Protocol and one Reactive Routing Protocol in a

mobile wireless network.

Analyze:

- ☐ **Route discovery process.**
- ☐ **Route maintenance mechanism.**
- ☐ **Routing overhead.**
- ☐ **Delay in establishing communication.**
- ☐ **Performance under high node mobility.**

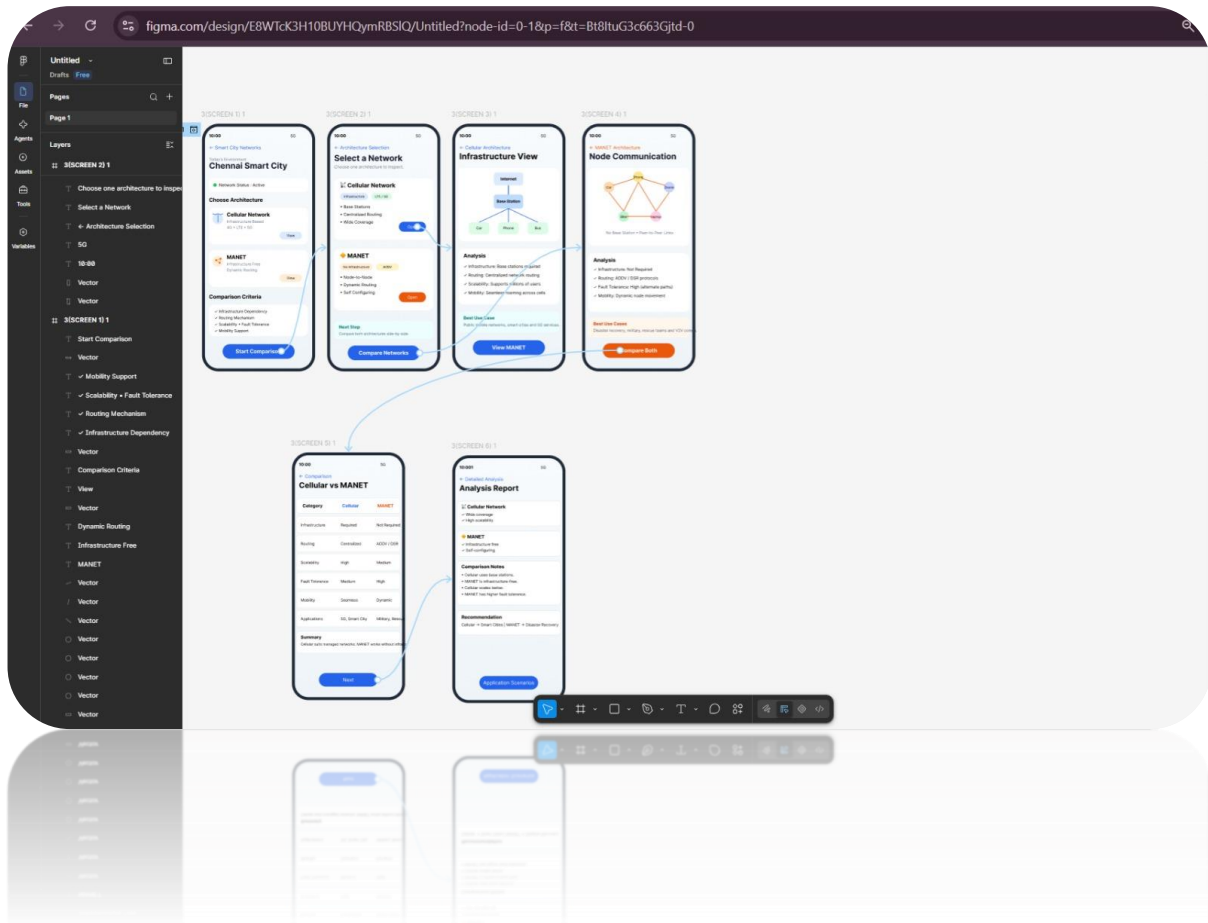
Include a comparative summary explaining which protocol performs better under different mobility

conditions.

Deliverables:

- ❑ Source code/design files.
- ❑ GitHub repository with implementation, screenshots, and analysis report.

PROTOTYPE:



FIGMA LNIK:

<https://www.figma.com/design/E8WTcK3H10BUYHQymRBSIQ/Untitled?node-id=0-1&p=f&t=Bt8ItuG3c663Gjtd-0>

1. Figma-Based Mobile IP Communication Workflow Analysis (BL4 – Analyze)

Design an interactive Figma prototype illustrating the complete workflow of Mobile IP communication

when a Mobile Node moves from its Home Network to a Foreign Network. Your design should include

the Home Agent (HA), Foreign Agent (FA), Correspondent Node (CN), Care-of Address (CoA),

Registration Process, and Packet Tunneling.

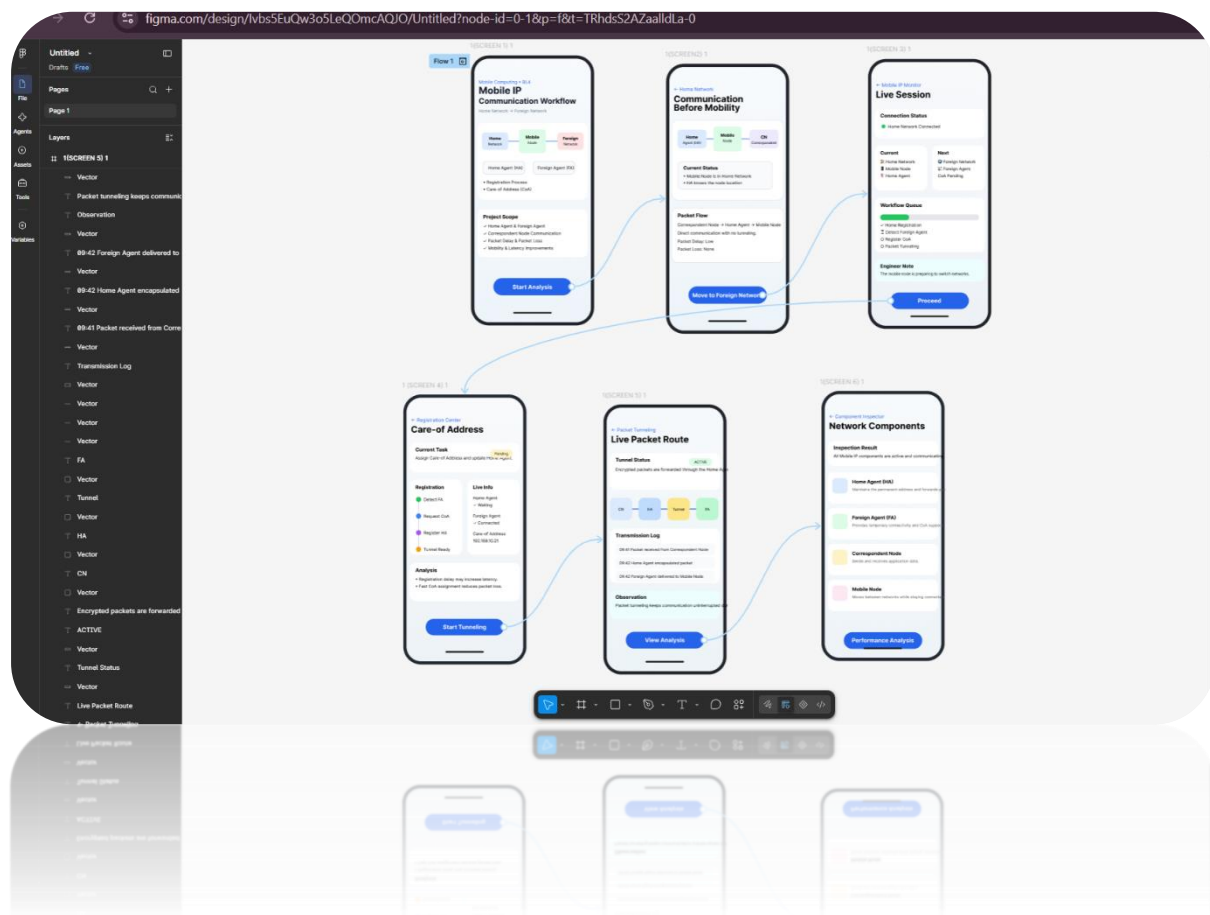
Analyze the communication process by identifying:

- ❑ The purpose of each network component.
- ❑ Points where packet delay or packet loss may occur.
- ❑ The effect of node mobility on communication performance.
- ❑ Possible improvements to reduce latency.

Deliverables:

- ❑ Figma (.fig) design
- ❑ GitHub repository containing exported images/PDF and documentation (README).

PROTOTYPE:



FIGMA LINK:

<https://www.figma.com/design/Ivbs5EuQw3o5LeQOmcAQJO/Untitled?node-id=0-1&p=f&t=TRhdsS2AZaalldLa-0>